

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

21st March 2017

Subject SA1 – Health and Care Insurance

Time allowed: Three hours (14.45* - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. * You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
- 3. You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
- 4. The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
- 5. Attempt all questions, beginning your answer to each question on a separate sheet.*
- 6. Mark allocations are shown in brackets.*
- 7. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** An Indian Standalone health insurance company that has been in existence for over 10 years is looking for business expansion. The product development team has proposed a new indemnity health product concept that incentivizes the insured for staying healthy.

The team intends to track policyholder's health by developing a technology based solution whereby policyholders will be asked to wear a device on their wrist. The device will capture multiple data points like step counts, heart rate, sleep pattern etc. and remotely send the data to the Insurance Company's server. The data will then be analyzed at regular intervals to decide on policyholder's health status. It is yet to be decided whether the insurer will manufacture its own device or partner with device manufacturers.

It has been proposed to offer discounts on premium at the time of renewal if the insured member has been in good state of health. This information will be used to decide on the levels of discount.

Premium shall not increase if the policyholder fails to follow health routine or takes initiatives to keep oneself healthy.

You are a pricing actuary and have been asked by appointed actuary to work on this proposal.

- i)** Discuss the proposed product design (5)
- ii)** Discuss the pricing approach with additional focus on the approach to calculate premium discounts and its structure. (12)
- iii)** Discuss the
 - a)** Underwriting and Sales (4)
 - b)** Administration and other considerations (4)
- iv)** Describe the following risks and suggest possible ways to mitigate them
 - a)** Mispricing risk (3)
 - b)** Reputational risk (3)
 - c)** Counterparty risk (assuming company decides to partner with device manufacturers) (3)
- v)** Evaluate whether the insurer should
 - a)** Manufacture the device themselves or acquire a company that manufactures such devices; (3)
 - b)** Partner with existing device manufacturers. (3)

The product team also wants to offer the proposed fitness based incentives to their existing policyholders of Indemnity plans to reduce anti-selective lapses and also to stand alone Critical Illness plans. Hence, they propose to offer it as a rider benefit which can be attached to existing policies at the time of renewal.

The Critical Illness plan is a standalone fixed benefit plan that offers lump sum benefits on first ever diagnosis / occurrence of any of the listed illnesses /procedures. The product offer covers against six illnesses including cancer and cardiovascular conditions.

- vi)** Evaluate the proposal (7)
- vii)** Discuss the approach to calculate rider premium. (3)

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Q. 2) A large overseas Health Insurance Company is considering to establish a Joint Venture Health Insurance Company in the Indian Market. The Company is also tracking the products, the regulatory environment, the distribution channels etc. currently prevailing within the Indian market.

- i)** List down and explain the 5 key points to be considered in choosing the local partner for the Joint Venture. (10)

A joint venture is established and the Company receives the regulatory approvals to start operations in India. While the company was going through the licensing process, the regulator also issued the “Guidelines on Product Filing in Health Insurance 2016”.

The Company’s decision for entering into the market was based on the existing practices and regulatory environment. You are an actuarial consultant in the Indian market and the foreign partner has approached you for your inputs.

- ii)** Prepare a note that covers the following: (3)
- a)** Key changes proposed in the guideline (3)
 - b)** The challenges and benefits it poses to the existing Health Insurers as well as new entrants in the market. (4)

Both the foreign and the Indian partners decided to go ahead with the JV and launched a number of products including Income protection (IP), Critical Illness (CI), Long Term Care (LTC) and PMI products which they serviced successfully for a few years making reasonable shareholder results on the overall portfolio basis. The Company then performed an experience analysis at a product level and realized that the Critical Illness products were turning out to be loss making whereas the income protection business was turning out to be very profitable.

It therefore proposed to stop selling Critical Illness product.

- iii)** Discuss the various potential implications of this proposal (8)

At the same time, given the success of the Income Protection product, the Company also proposed to launch a micro insurance Income Protection product (low premiums and low benefit limits)

- iv)** Discuss how the experience under the micro insurance product may differ than that from the standard IP product and suggest possible ways of improving the experience under the micro-insurance IP Product. (13)

To further enhance the suite of products, the company also launched a new PMI product where the initial Sum Insured (100%) is automatically reinstated in the same policy year as soon as the initial Sum Insured is fully consumed.

- v)** Explain the conditions the Company should apply to manage the benefit provided and mitigate any risk associated with providing this benefit. (4)
- vi)** How will you determine the cost of providing the Sum Insured reinstatement benefit? (8)

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